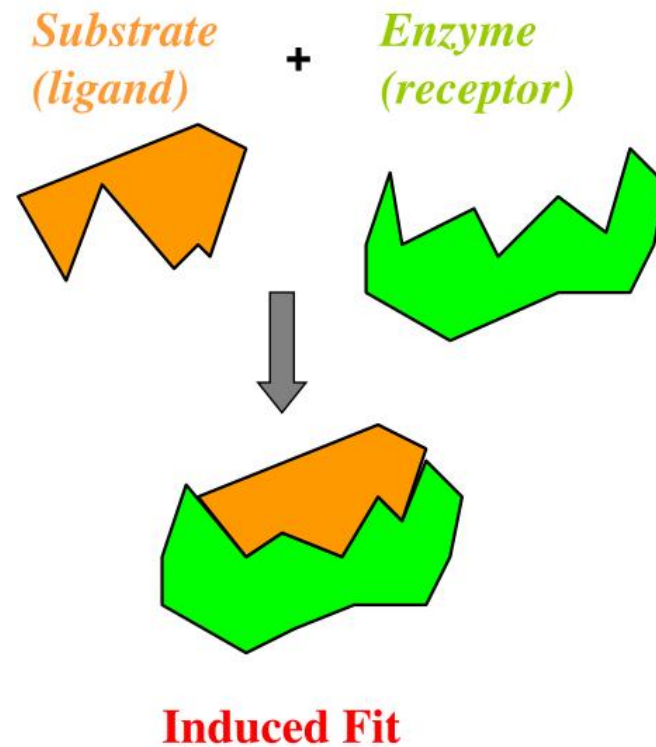
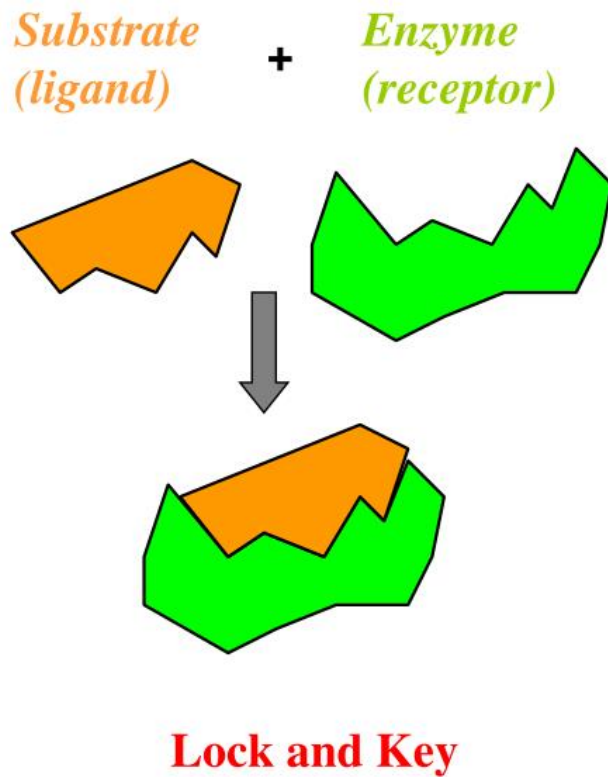


Quiz on Monday!

Target: Protein (basics)

Complementarity in substrate-enzyme interaction



Docking

Rigid Docking

- This approximation treats both the ligand and the receptor as rigid and

Flexible Docking

- A more common approach is to model the ligand flexibility while

In biology, interactions are dynamic

ligand binding), are dealt with by rigid protein docking.

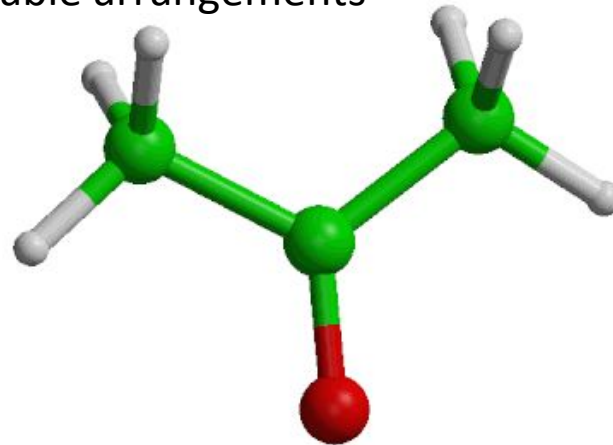
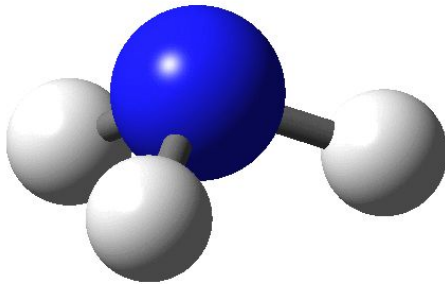
- Computationally inexpensive

induced fit docking is used with flexible binding sites

- Computationally expensive

Atoms never stop jiggling

- In real life, atoms are in constant motion.
 - They will *not* go to an energy minimum and stay there.
- Given enough time, the simulation samples the Boltzmann distribution
 - That is, the probability of observing a particular arrangement of atoms is a function of the potential energy
 - In reality, one often does not simulate long enough to reach all energetically favorable arrangements

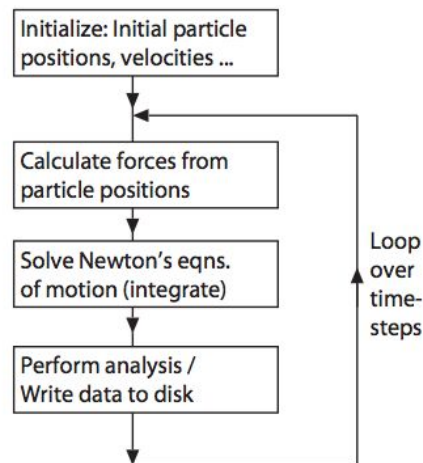


Capturing the dynamics (MD simulation)

- Simulation of explicit particles (atoms, ions, . . .)
- Particles interact via relatively simple analytical potential functions
- Newton's equations of motion are integrated for all particles simultaneously
- 100-1,000,000's of particles depending on model

Steps in Performing an MD Simulation

- ▶ Selection of interaction model
- ▶ Selection of boundary conditions
- ▶ Selection of initial conditions (positions, velocities ...)
- ▶ Selection of ensemble (NVE, NVT, NPT ...)
- ▶ Selection of target temperature, density/pressure ...
- ▶ Selection of integrator, thermostat, barostat ...
- ▶ Perform simulation until equilibration is reached (property dependent)
- ▶ Perform production simulation to collect thermodynamic averages, positions, velocities
- ▶ Analyze the results via post-processing



Tools:
GROMACS,
AMBER, NAMD
etc.

To be taught in
subsequent weeks!

Solving the equations of motion



$$U(\vec{R}) = \sum_i k_i^{bond} (r_i - r_0)^2 + \sum_i k_i^{angle} (\theta_i - \theta_0)^2 +$$

MD Simulation: To capture fluctuations and conformational changes of proteins and stability of protein-ligand interaction

version number

$U_{nonbond}$

U_{bond} = oscillations about the equilibrium bond length

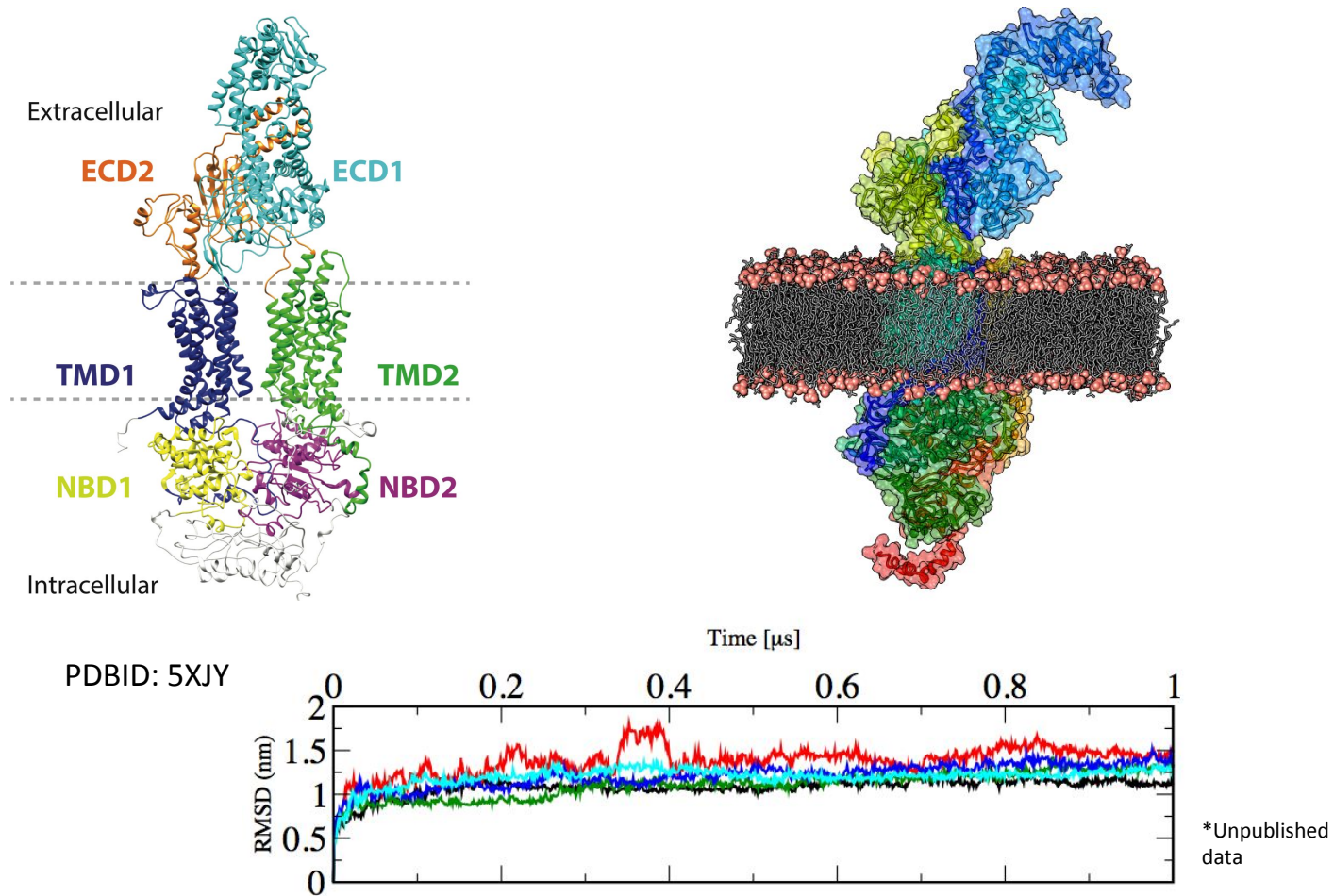
U_{angle} = oscillations of 3 atoms about an equilibrium bond angle

$U_{dihedral}$ = torsional rotation of 4 atoms about a central bond

$U_{nonbond}$ = non-bonded energy terms (electrostatics and Lenard-Jones)

ROLE OF ENVIRONMENT

Solving the equations of motion



Environment matters!

Starting crystal structure

End structure (1000ns)

1.83

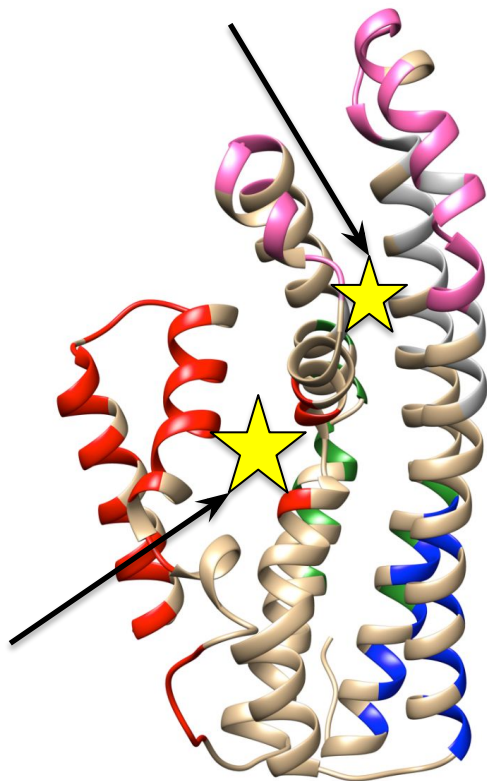
*Unpublished
data

models	molpdf	DOPE score	GA341 score
1XYZ.B99990001.pdb	1185.43982	-31210.45871	1.00000
1XYZ.B99990002.pdb	1101.32812	-31389.98212	1.00000
1XYZ.B99990003.pdb	1079.27711	-31455.78214	1.00000
1XYZ.B99990004.pdb	1052.94351	-31290.14389	1.00000
1XYZ.B99990005.pdb	1132.56111	-31567.00984	1.00000

The “Door Paradigm”



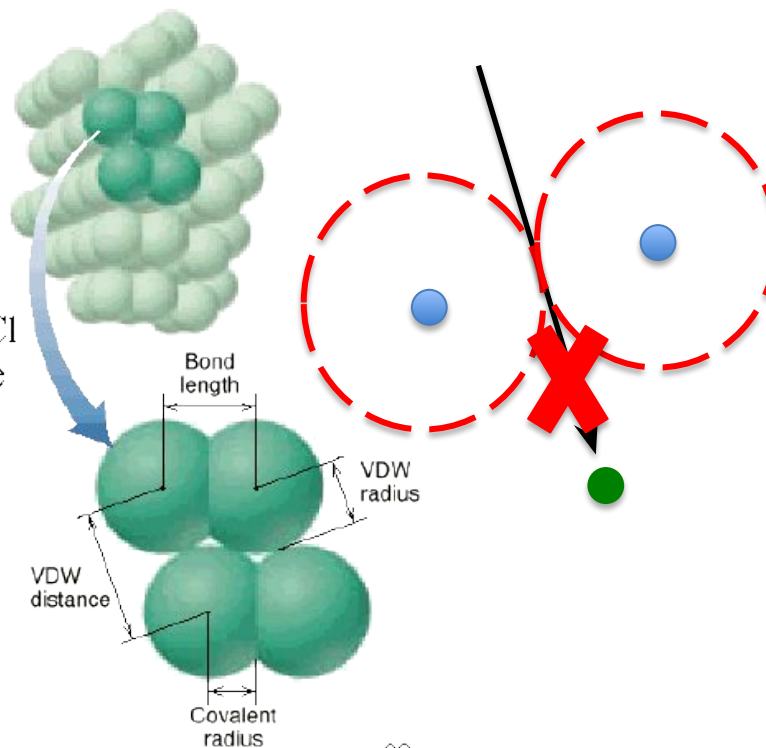
The “Door Paradigm”



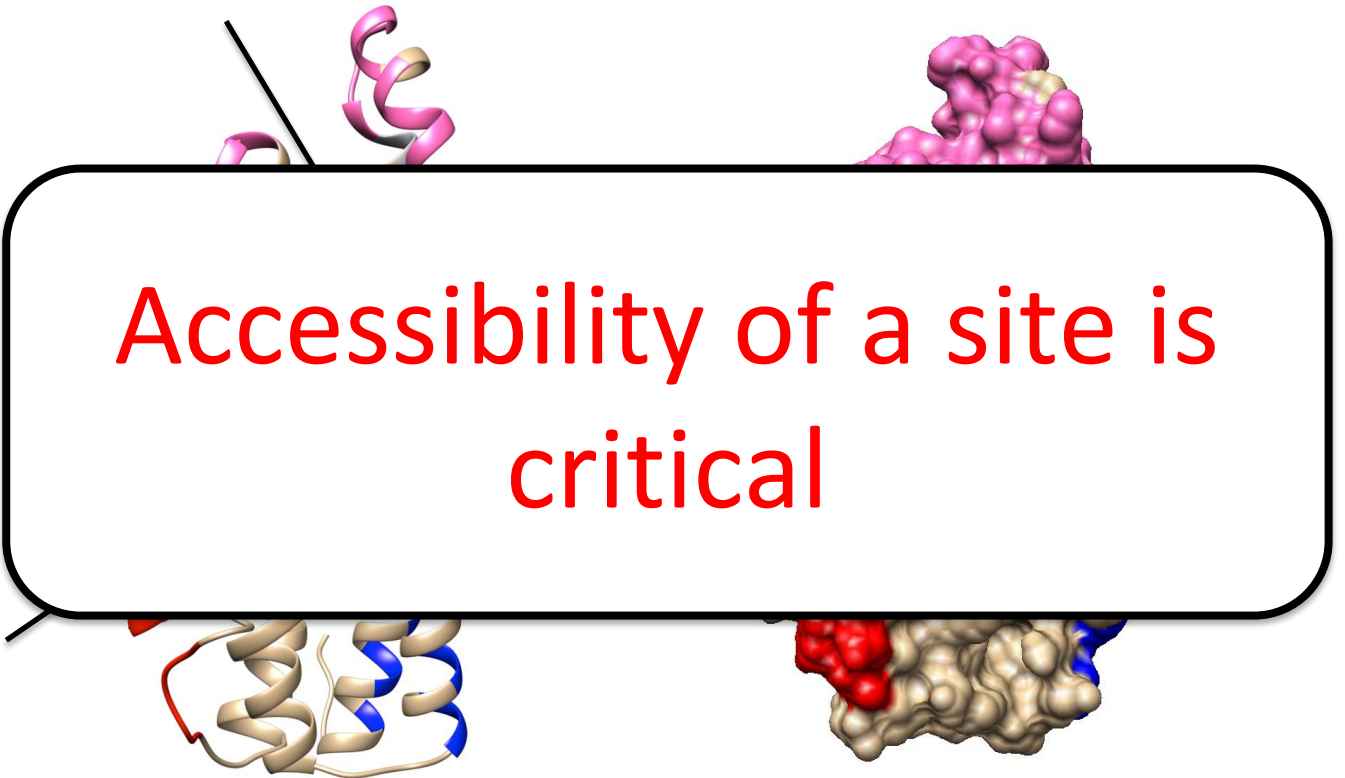
Ray, A., Ghosh, A., Chakraborty, R., Upadhyay, S.K., Maiti, S., Sengupta, S. and Thukral, L., 2018. Specific cholesterol binding drives drastic structural alterations in Apolipoprotein A1. *The journal of physical chemistry letters*, 9(20), pp.6060-6065.

The “Door Paradigm”

- The distance between two Cl atoms in the Cl_2 molecule is the bond length; one-half this distance is the covalent radius.
- The distance between two nonbonded Cl atoms in adjacent molecules is called the **van der Waals distance**; one half this distance is the VDW radius.
- The covalent radius is always less than the van der Waals radius.



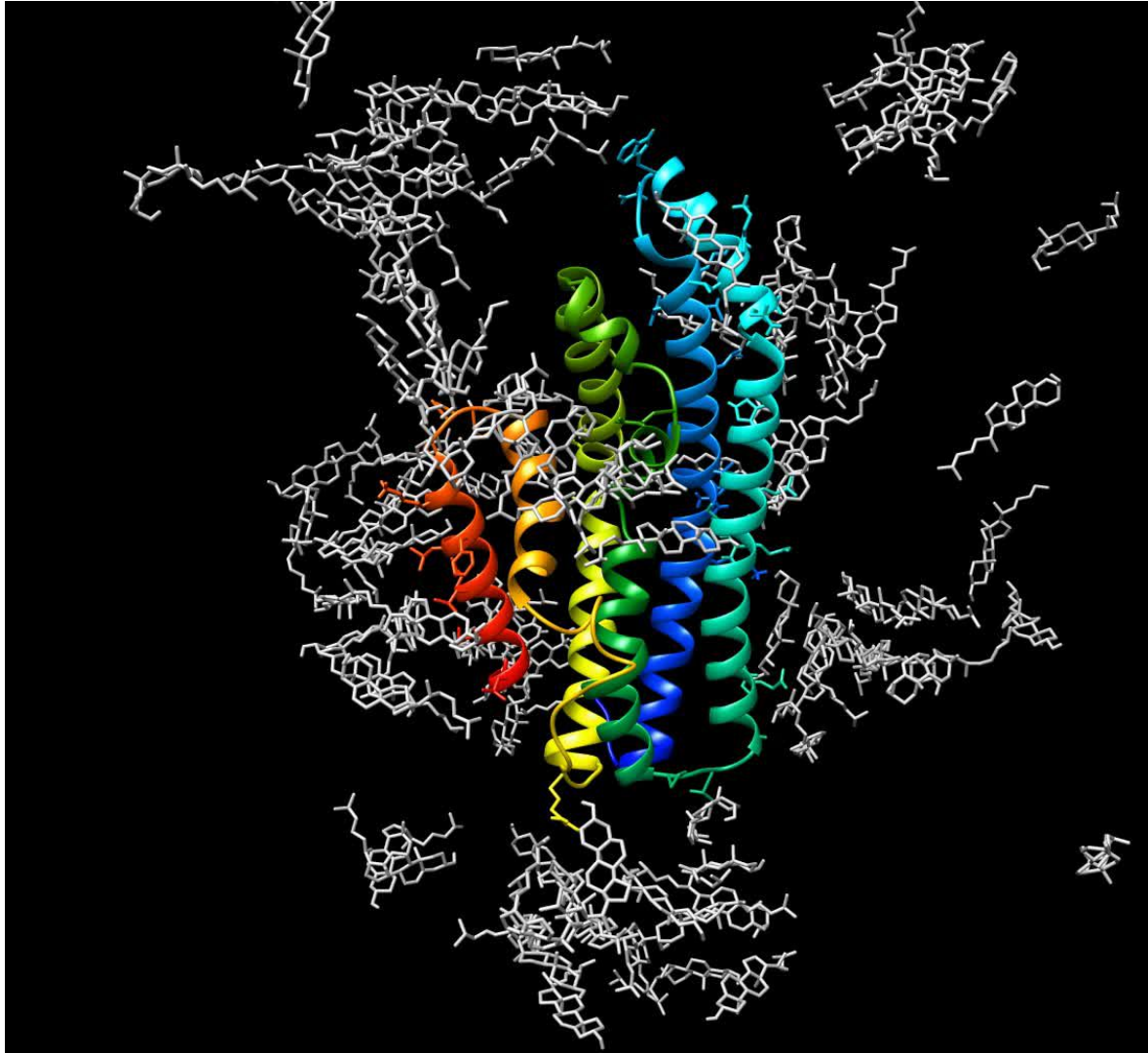
The “Door Paradigm”

The image features a central white rounded rectangle with a black border containing the text 'Accessibility of a site is critical' in red. Surrounding this rectangle are four molecular models of Apolipoprotein A1 (ApoA1). At the top left, a pink ribbon model is shown with a black line indicating a specific site. At the top right, a pink surface representation of the protein is visible. At the bottom left, a ribbon model shows a transition state with a red segment and a blue segment. At the bottom right, a surface representation shows a red segment and a blue segment, likely representing different states of the protein's 'door' region.

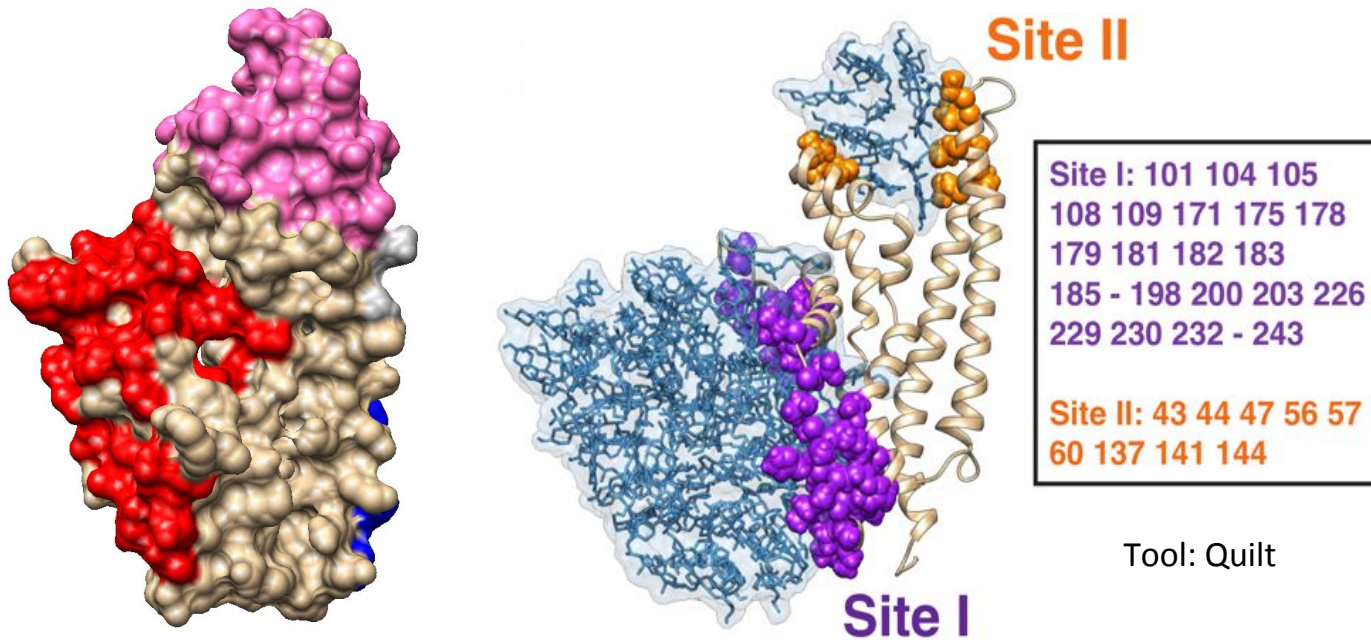
Accessibility of a site is
critical

Ray, A., Ghosh, A., Chakraborty, R., Upadhyay, S.K., Maiti, S., Sengupta, S. and Thukral, L., 2018. Specific cholesterol binding drives drastic structural alterations in Apolipoprotein A1. *The journal of physical chemistry letters*, 9(20), pp.6060-6065.

AN EXTREME CASE OF PROTEIN-LIGAND INTERACTION

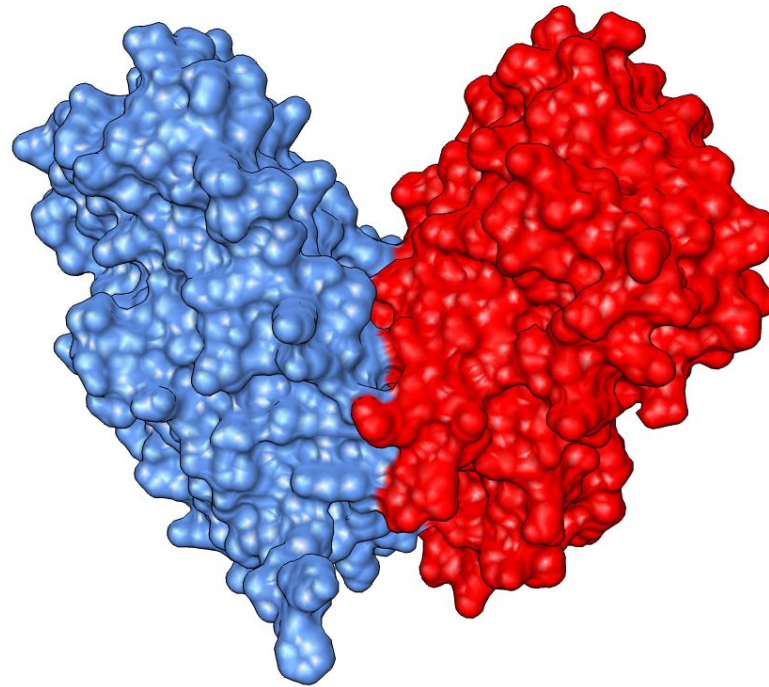


Hydrophobic interaction



Ray, A., Ghosh, A., Chakraborty, R., Upadhyay, S.K., Maiti, S., Sengupta, S. and Thukral, L., 2018. Specific cholesterol binding drives drastic structural alterations in Apolipoprotein A1. *The journal of physical chemistry letters*, 9(20), pp.6060-6065.

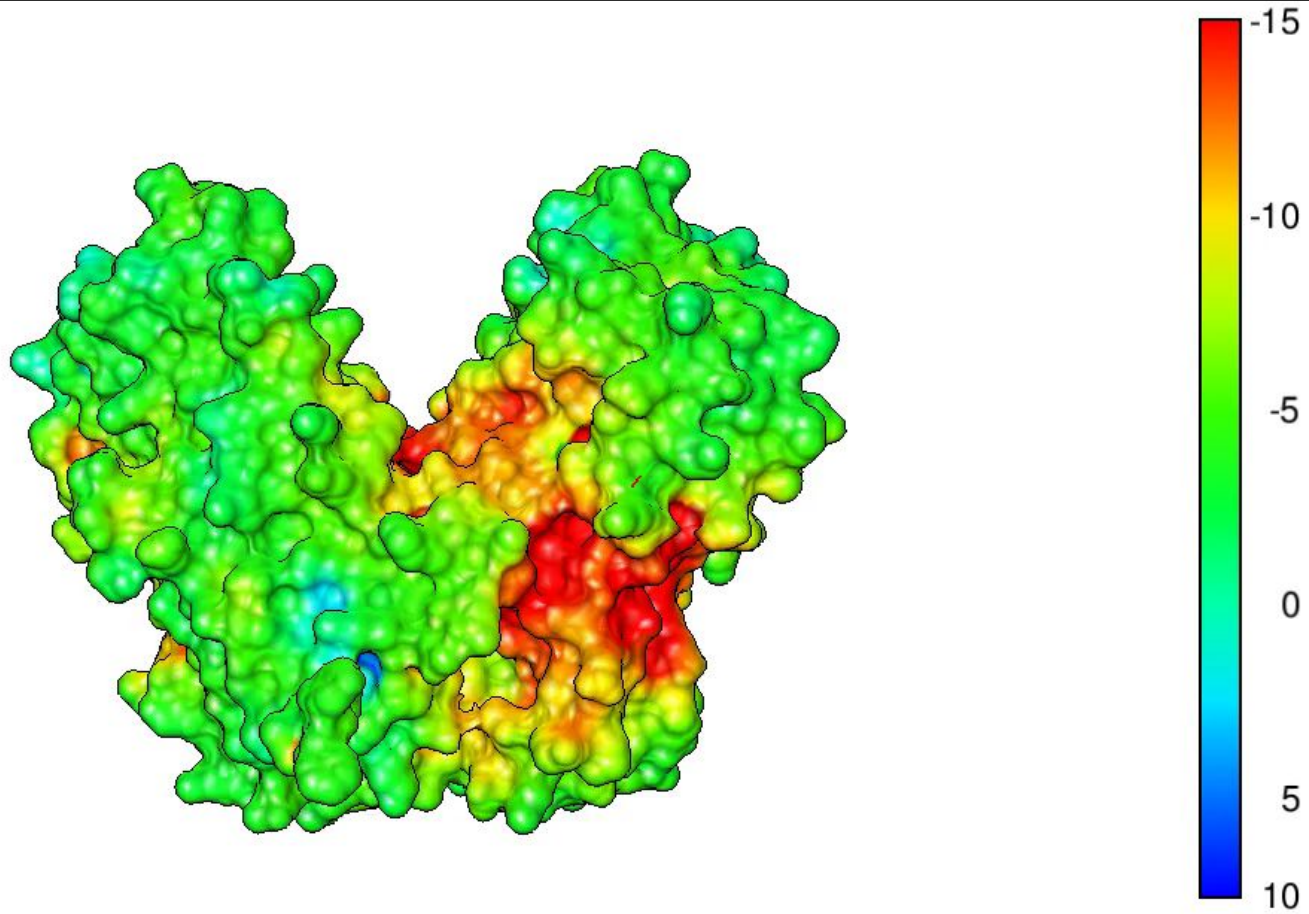
Electrostatics



PDBID: 4UTP

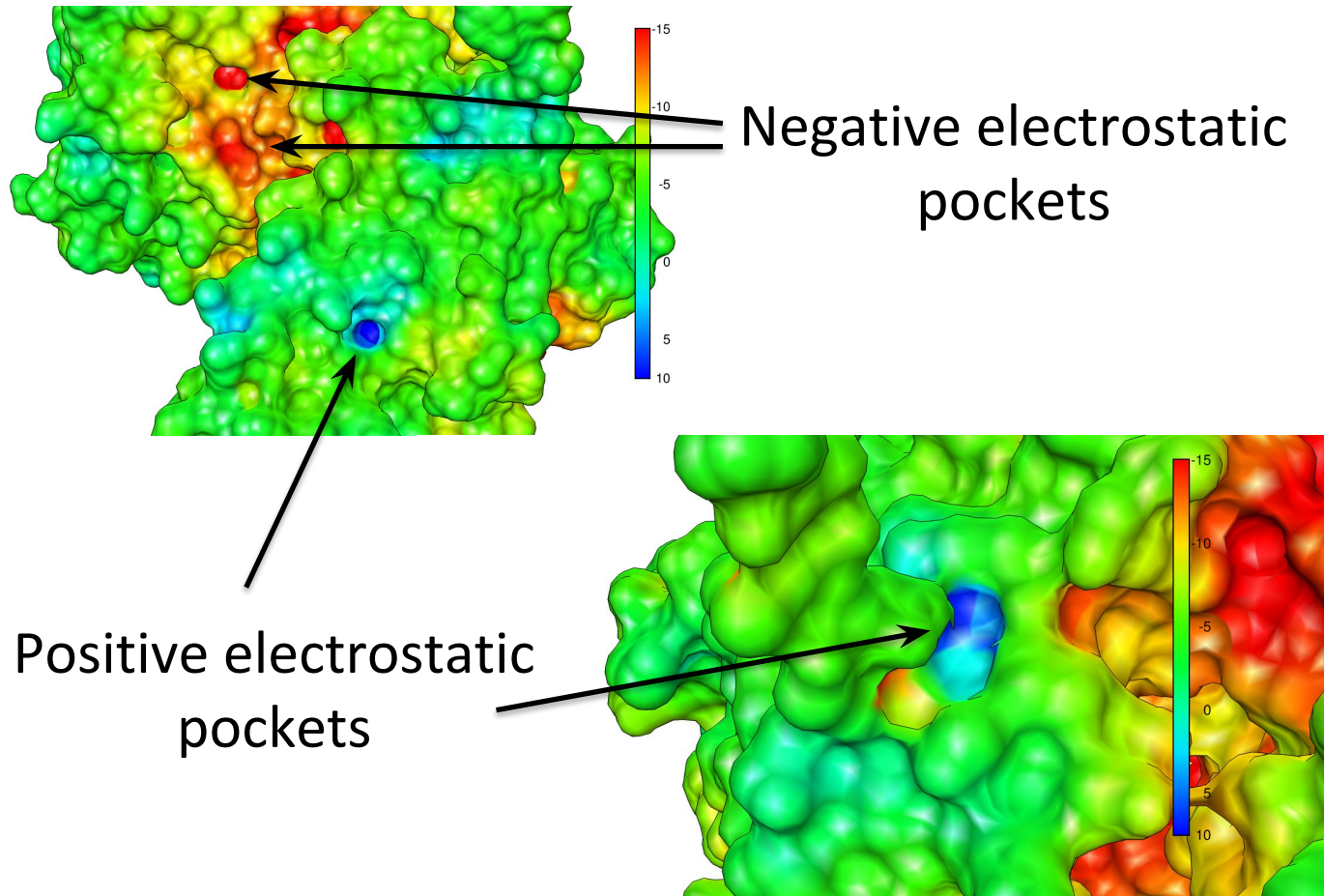
Crystal structure of pneumococcal surface antigen PsaA

Electrostatics

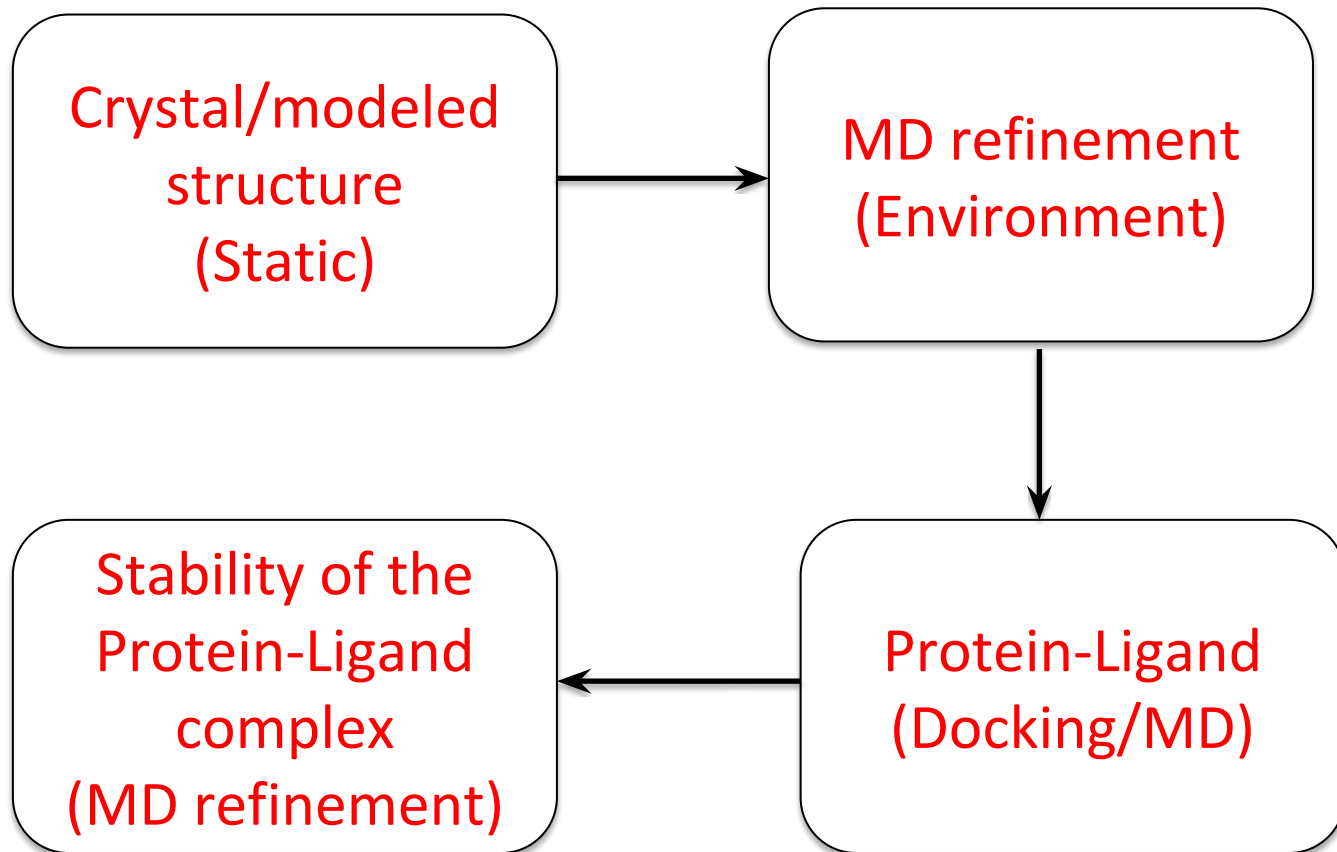


Tool: pdb2pqr,
apbs

Electrostatics



Workflow to consider



Recap

- Introduction to three-dimensional structure of protein
- Homology modeling
- Molecular dynamics
- Environment effect
- Accessibility of sites